



International Transaction Journal of Engineering, Management, & Applied Sciences & Technologies

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Smart and Green Amra City: An IoT-Driven Urban Framework for Sustainable City Development and Intelligent Infrastructure in Jordan

Saif Mutlq^{1*}, Deema Nmour²

¹ Department of Computer Engineering, Jordan University of Science and Technology (JUST), Irbid, Jordan.

² Department of Computer Engineering, Jordan University of Science and Technology (JUST), Irbid, Jordan.

*Corresponding Author (Email: smmetlaq23@cit.just.edu.jo, dialnmour23@cit.just.edu.jo)

Keywords:

Smart City; Internet of Things (IoT); Green Urbanism; Edge Computing; Fog Computing; Digital Twin; City Operating System (CityOS); Structural Health Monitoring (SHM); Sustainable Urban Development; Private 5G Networks; Sponge City; Mobility-as-a-Service (MaaS); Zero-Trust Security; Smart Stadium; Arid Urban Development.

Abstract

By 2050, nearly 7 billion people are expected to live in urban areas, placing unprecedented pressure on resources, infrastructure, and data systems. In Jordan, rapid population growth in cities such as Amman and Zarqa highlights the limitations of conventional urban planning approaches. This paper presents Amra City, a 500 km² smart and sustainable urban development designed to address these challenges through an integrated, technology-driven framework. The proposed architecture combines IoT-based sensing networks, edge and fog computing, private 5G connectivity, and a 4D Digital Twin, all orchestrated through a unified City Operating System (CityOS). A key contribution is the role of the international stadium as a multifunctional smart infrastructure asset: a net energy producer, rainwater harvesting hub, structural health monitoring testbed, and resilient event operations center. The framework integrates Green IoT principles, Sponge City water strategies, and Mobility-as-a-Service (MaaS) to enhance sustainability, efficiency, and adaptability in arid environments. A comprehensive cybersecurity architecture and privacy-by-design approach ensures secure city-wide operations. This work provides a scalable, replicable blueprint for sustainable smart urban transformation in resource-constrained environments.

Disciplinary: Smart City Engineering, IoT Systems, Urban Planning, Sustainable Development, Cybersecurity.

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Cite This Article:

Mutlq, S., and Nmour, D. (2026). Smart and Green Amra City: An IoT-Driven Urban Framework for Sustainable City Development and Intelligent Infrastructure in Jordan. International Transaction Journal of Engineering, Management, & Applied Sciences & Technologies, 20([Issue]), [ID], 1-12. [http://TUENGR.COM/V16/\[ID\].pdf](http://TUENGR.COM/V16/[ID].pdf) DOI: 10.14456/ITJEMAST.2025.

1 Introduction

The world is urbanizing at an unprecedented rate, reshaping the relationship between populations, infrastructure, and the environment. As cities expand under the pressures of population growth, climate change, and digital transformation, the limitations of conventional urban planning approaches are becoming increasingly evident —particularly in resource-constrained regions. Jordan presents a clear example of these challenges. Rapid urban concentration in Amman, combined with water scarcity and sustained population growth, has placed significant strain on existing infrastructure systems. These conditions highlight the need for new urban development paradigms that prioritize efficiency, resilience, and sustainability. In response, Amra City has been proposed as a large-scale smart and sustainable urban development near Amman. Spanning approximately 500,000 dunams, the city is designed to accommodate over one million residents while reducing pressure on the capital and supporting economic growth through structured investment frameworks. Rather than focusing solely on scale, Amra emphasizes integrated intelligence as a core design principle. At the core of Amra's vision is the seamless integration of Internet of Things (IoT) technologies, green infrastructure, and data-driven urban management systems. Digital intelligence is embedded across key urban domains, including energy, water, mobility, and public services. A central element of this approach is the international stadium, which serves as a multifunctional smart infrastructure asset and a practical testbed for the city's broader technological ecosystem.

Despite the growing body of research on smart city frameworks, a critical gap remains. Most existing models—such as Songdo's sensor-intensive infrastructure, Barcelona's data-driven urban management, and Singapore's integrated platforms—have been developed in resource-rich environments with mature digital ecosystems. As a result, their direct applicability to cities facing water scarcity, limited public resources, and rapid population growth remains constrained. In addition, many implementations focus on individual urban domains in isolation, lacking a unified framework that integrates IoT infrastructure, environmental sustainability, and large-scale operational systems within a cohesive architecture. This paper addresses these limitations by presenting Amra City as a comprehensive, context-aware urban model that aligns technological innovation with environmental and economic constraints. The proposed framework offers a scalable and adaptable approach for smart urban development in resource-constrained settings.

Against this backdrop, this paper addresses the following research question:

How can an integrated IoT-driven urban framework be designed to simultaneously achieve environmental sustainability, operational resilience, and large-scale event readiness within a resource-constrained arid-region context?

Three subsidiary questions guide the framework development: (1) What computing and sensing architecture enables real-time urban intelligence while maintaining operational autonomy independent of cloud connectivity? (2) How can a single infrastructure asset — the international stadium — be designed to serve as a net contributor across energy, water, structural monitoring, and event management domains simultaneously? (3) What cybersecurity posture is appropriate for a hyper-connected urban environment where thousands of heterogeneous IoT devices govern safety-critical systems? This paper is structured as follows: Section 2 reviews related work and global smart city initiatives. Section 3 introduces the Amra City vision and urban context.

Sections 4 and 5 describe the system components and adaptability framework. Sections 6 and 7

present the smart city architecture and communication technologies. Sections 8 and 9 examine edge and fog computing applications, while Section 10 details the Digital Twin framework. Section 11 discusses cybersecurity and data protection, followed by conclusions and future implications.

1.1 Research Methodology

This paper adopts a Design Science Research (DSR) methodology, as formalized by Hevner et al. (2004), to develop and evaluate the Amra City framework as a sociotechnical artifact addressing the urban challenges of rapid population growth, water scarcity, and digital infrastructure gaps in Jordan. The DSR process followed three iterative phases.

In the problem identification phase, urban challenges specific to Jordan's context were systematically characterized through analysis of national policy documents — including Jordan's Economic Modernization Vision (2022) and the National Water Strategy 2023–2040 — alongside a structured review of existing smart city literature spanning 2014–2024. This review covered IoT infrastructure models, edge computing deployments, digital twin implementations, and sustainability frameworks, identifying the critical gap: the absence of a unified, resource-constrained smart city architecture that simultaneously integrates IoT sensing, environmental sustainability, large-scale event operations, and cybersecurity.

In the design and development phase, the Amra City framework was constructed through comparative analysis of global smart city precedents — including Songdo, Barcelona, Singapore, and Copenhagen — evaluated against Jordan's specific environmental, economic, and infrastructural constraints. Technology selection decisions (network protocols, computing tiers, sensing modalities) were grounded in published technical benchmarks and peer-reviewed performance data.

In the evaluation phase, the framework's components were assessed against established smart city performance criteria, including latency thresholds, energy efficiency targets, water conservation benchmarks, and cybersecurity standards, using quantitative estimates derived from comparable deployed systems and validated environmental datasets for the Amman region.

2 Background & Related Work

The evolution of smart cities over the past two decades has established a substantial body of knowledge upon which Amra City builds. Early implementations demonstrated the transformative potential of IoT-driven urban management. Songdo International Business District in South Korea deployed extensive sensor networks across its urban infrastructure, enabling automated waste collection through pneumatic systems and real-time energy orchestration, with reported efficiency improvements of up to 25%. Barcelona's 22@ innovation district integrated air quality monitoring networks that significantly reduced environmental threshold alerts, while Copenhagen's Vehicle-to-Everything (V2X) communication infrastructure contributed to measurable reductions in traffic-related incidents. These pioneering deployments collectively demonstrated that IoT-enabled cities can process large-scale data streams across interconnected urban domains, establishing foundational principles that inform Amra's architectural approach. Beyond connectivity, a second generation of smart city initiatives shifted toward environmental sustainability, giving rise to the Green IoT

paradigm. Amsterdam's circular waste management system leveraged dynamic routing algorithms to significantly reduce fuel consumption, while Helsinki deployed occupancy-adaptive street lighting systems achieving substantial energy savings. In the water domain, Dubai's controlled-environment agriculture initiatives demonstrated considerable reductions in water consumption through precision IoT-driven irrigation. At the regional level, Amman's Green City Action Plan introduced IoT-enabled composting and urban greening pilots, providing a locally relevant precedent for integrating environmental monitoring with city management. Collectively, these initiatives highlight the potential of low-power wide-area networks (LPWAN) and energy-efficient sensing protocols to support sustainability goals in resource-constrained environments—a challenge directly aligned with Jordan's urban context. In parallel, advances in ICT infrastructure have enabled the interoperability and scalability required for city-wide smart systems. Shenzhen's large-scale deployment of smart metering infrastructure demonstrated high system reliability in dense urban environments. Dubai's adoption of FIWARE-based open platforms enabled interoperability across hundreds of applications, establishing a replicable model for vendor-neutral urban data ecosystems. Within Jordan, the Amman Smart City Roadmap has laid foundational groundwork through traffic management systems and e-governance initiatives, supported by public–private partnerships and expanding digital infrastructure. These developments provide a technical and institutional baseline upon which more advanced smart city architectures, such as Amra, can be developed.

Despite these advances, a consistent set of limitations persists across existing smart city models. Cloud-centric architectures, which underpin many deployed systems, introduce non-negligible latency that can become critical in high-density scenarios such as large-scale events or emergency response operations. Studies indicate that large-scale smart city deployments may experience performance degradation under peak load conditions, particularly when centralized architectures are heavily relied upon. Furthermore, many existing frameworks were designed for cities with stable resource availability and mature digital ecosystems, limiting their direct applicability to arid and rapidly urbanizing environments. Critically, few models provide a unified framework that simultaneously integrates IoT-driven infrastructure, environmental sustainability systems, large-scale operational management, and cybersecurity within a single coherent architecture tailored to resource-constrained contexts. It is this gap that the Amra City framework is designed to address.

3 Green Amra City Vision and Urban Context

The Green Amra City vision is grounded in the Green City paradigm, which prioritizes functional sustainability over aesthetics, emphasizing optimized resource utilization, waste reduction, and environmental resilience. Amra is conceptualized as an integrated urban system aligned with the Triple Bottom Line framework, addressing environmental performance, social well-being, and economic efficiency in a unified manner. Environmentally, the city focuses on energy efficiency, renewable energy integration, and climate-responsive urban design. Socially, it promotes healthier living conditions, accessible public spaces, and reduced pollution levels. Economically, smart infrastructure and efficient resource management deliver long-term operational savings, particularly within the energy–water nexus — an essential consideration in arid regions

such as Jordan. Given Jordan's chronic water scarcity, water-sensitive urban design represents a central pillar of Amra's master plan. The project adopts the Sponge City concept, which aims to capture, store, and reuse rainwater rather than allowing it to be lost as surface runoff. While cities such as Shenzhen and Singapore are recognized for their advanced digital infrastructure, their application of Sponge City principles offers relevant insights for water resource optimization in arid environments, where rainfall is limited but the value of captured water is substantial. Within this framework, the international stadium serves as a key infrastructural component — its large roof surface and surrounding permeable public spaces enable large-scale rainwater harvesting, with collected water directed into underground storage systems, significantly offsetting irrigation demands and reducing reliance on groundwater resources.

The implementation of Green Amra City integrates both technical and non-technical dimensions. On the technical side, Green IoT solutions enable real-time monitoring and optimization of energy and water systems through smart grids, environmental sensing, and distributed renewable energy installations. On the urban design side, passive strategies — including building orientation, wind corridors, walkability, and transit-oriented development — reduce energy demand and mitigate the Urban Heat Island effect. Through this integrated approach, Amra City moves beyond conventional urban development models, presenting a context-aware framework for sustainable desert urbanism.

4 Characteristics and Core Components of Amra City

The architectural and operational framework of Amra City is defined by the integration of physical infrastructure with digital intelligence. Unlike conventional urban developments, Amra is structured around five core components that collectively enable a responsive, resilient, and sustainability-oriented urban ecosystem. These components support environmental performance, operational efficiency, and quality of life, while positioning the international stadium as an integral smart urban asset rather than an isolated facility.

-Urban Planning Components: Amra City adopts the 15-Minute City planning model, ensuring that residents can access essential services—including healthcare, retail, education, green spaces, and public transport—within a 15-minute walking or cycling distance. This concept extends to the stadium district, where the urban layout supports high-density event scenarios by enabling efficient access to accommodation, mobility hubs, and services. By reducing reliance on long-distance travel, this approach contributes to lower congestion levels, reduced emissions, and improved urban accessibility.

-Green Infrastructure: Green infrastructure in Amra City is designed as a functional component of the urban system. Multi-purpose green spaces incorporate bio-swales for stormwater management, water quality improvement, and localized cooling. Permeable pavements enable rainwater infiltration and support groundwater recharge, reinforcing the Sponge City strategy introduced at the neighborhood scale.

-Smart and Green Mobility: Amra City implements a Smart and Green Mobility framework based on the Mobility-as-a-Service (MaaS) model, integrating multiple transportation modes within a unified digital platform, including public transit, shared mobility, pedestrian pathways, and future mobility services. In high-density pedestrian zones, particularly around the stadium, energy-harvesting walkways are proposed to capture mechanical energy from foot traffic. While such technologies are still emerging, they demonstrate the potential for embedding localized energy generation within urban mobility infrastructure.

-Waste and Water Management: Amra City adopts a circular resource management approach, treating waste streams as recoverable resources. A pneumatic waste collection system is proposed to transport waste from distributed smart bins to centralized processing facilities, reducing reliance on conventional collection vehicles and improving urban hygiene. In parallel, a smart water distribution network equipped with sensing technologies enables continuous monitoring and early leak detection—an essential feature in water-scarce environments where minimizing losses is critical.

-International Stadium as a Smart Infrastructure Component: The international stadium functions as an integrated component within Amra's urban system. It is designed to support on-site renewable energy generation through large-scale solar installations that can contribute to the surrounding energy network. In addition, a Structural Health Monitoring (SHM) system—utilizing embedded sensors such as strain gauges and accelerometers—enables continuous assessment of structural performance. These data streams can be integrated into a digital twin environment to support predictive maintenance and operational safety. Coordinated through the City Operating System (CityOS), the stadium interacts with surrounding urban systems, including mobility, energy, and services.



Figure 1: Stadium-centric smart infrastructure of Amra

5 Adaptability Framework for Green and Modern City Requirements

The long-term value of Amra City lies in its adaptability —its capacity to respond effectively to environmental challenges, technological evolution, and societal needs. Beyond achieving smart functionality, adaptability ensures urban resilience, operational continuity, and sustainability over time. This section evaluates how Amra City aligns with contemporary green and modern city requirements across four dimensions: environmental, technological, social, and event-based adaptability, positioning the city as a scalable and future-ready urban model.

5.1 Environmental Adaptability and GCAP Integration

Amra City achieves strong environmental adaptability by integrating climate-responsive design principles aligned with national and municipal sustainability strategies, including frameworks consistent with the Amman Green City Action Plan (GCAP). Given Jordan's water scarcity and rising urban temperatures, the city incorporates adaptive water and cooling solutions designed to minimize resource consumption under varying climatic conditions. A proposed closed-loop smart cooling system utilizes recovered HVAC condensation from the international stadium, combined with treated greywater, to support localized cooling zones in high-density public areas. These zones are activated through environmental and occupancy sensors during peak heat or crowd concentration periods. Building upon the Urban Heat Island mitigation strategies introduced in Section 4, this system operationalizes those principles at the district scale—translating passive design intent into active, sensor-driven climate response without imposing additional strain on freshwater resources.

5.2 Technological Adaptability

Technological adaptability is a critical determinant of smart city resilience, particularly in a national context like Jordan's, where public investment cycles and infrastructure procurement must balance ambition with fiscal pragmatism. Amra City is designed with a modular and upgradable digital infrastructure, allowing seamless integration of emerging technologies without disrupting existing services. Open standards, interoperable platforms, and software-defined networking enable the city to evolve alongside advancements in IoT, communication technologies, and intelligent computing paradigms. This flexibility ensures that Amra City avoids technological lock-in and system obsolescence—a risk observed in several smart city implementations. By maintaining continuous technological readiness, the city can incorporate future innovations such as advanced AI analytics, next-generation connectivity, or autonomous systems, while preserving operational stability and protect prior infrastructure investment.

5.3 Social and Health Adaptability

Amra City adopts a human-centric approach by embedding health and IoT based social considerations within its digital infrastructure. The city aligns with international healthy city principles, including guidelines established by the World Health Organization (WHO), to promote physical and mental well-being across diverse population groups. An Integrated Health Alert System leverages real-time air quality and environmental sensors to notify residents and visitors through a city-wide mobile platform when pollutant levels exceed recommended thresholds. Alerts are tailored for vulnerable groups, such as individuals with respiratory conditions, ensuring proactive risk

mitigation. Social adaptability is further reinforced through digital inclusive measures, including widespread high-speed connectivity in public spaces, enabling equitable access to digital services and fostering social cohesion across Amra’s diverse residents and visitor populations.

5.4 City Readiness for Large Smart Events

The adaptability of Amra City is critically tested during large-scale events hosted at the international stadium. To manage extreme fluctuations in population density and data traffic, the city operates an Event Mode framework that synchronizes urban systems in real time—transforming the city’s baseline operational state into a dynamic, high-capacity event configuration. Crowd management within this framework relies on AI-driven analytics integrated with the stadium’s digital twin, enabling continuous simulation of crowd dynamics and early detection of congestion risks. When predefined thresholds are exceeded, automated response mechanisms activate dynamic signage, mobility rerouting, and mobile alerts to redistribute crowds safely. Simultaneously, smart grid stability is maintained through the stadium’s Battery Energy Storage Systems (BESS), absorbing peak demand surges and preventing stress on the national grid. Coordinated through the CityOS, these mechanisms ensure public safety, operational reliability, and seamless urban experience during large-scale events. The technical underpinning of these capabilities—spanning edge computing, IoT sensing, and digital twin analytics—is examined in detail in the sections that follow.



Figure 2: Adaptability Framework of Amra City

6 Smart City Architecture for Amra

Smart city architecture provides the structural blueprint that links urban infrastructure with digital intelligence, defining how data flows from physical assets to decision-making systems and citizen services. Building on the global precedents examined in Section 2, Amra’s architectural backbone is not a technical afterthought but the central mechanism through which its Green City vision, IoT-driven sustainability, and event-ready stadium ecosystem become operational realities—enabling the CityOS, Green IoT strategies, and Event Mode framework described in earlier sections to function as an integrated whole.

6.1 Smart City Architecture Layers

The architecture of Amra Smart City is organized into four interdependent layers—Sensing, Network, Data Processing, and Application—mirroring global best practices while being tailored to Jordan’s environmental constraints and Amra’s role as a smart-event hub. This layered structure ensures that the international stadium, Green IoT systems, smart mobility, and circular resource management operate on a common digital foundation rather than as isolated projects.

The **Sensing layer** represents the city's digital nervous system, featuring multi-modal sensors embedded directly into infrastructure rather than standalone devices. These include strain gauges and accelerometers embedded in stadium concrete to support long-term structural health monitoring over several decades, AI-powered cameras for crowd analytics, environmental sensors for air quality and water conditions, and acoustic sensing via underground fiber-optic cables (distributed acoustic sensing) for high-precision leak detection and movement monitoring. Edge-enabled intelligence in these sensing nodes enables localized, real-time decisions, such as automatically shutting valves during major leaks, drawing inspiration from resilient designs in advanced smart city deployments to support Amra's Sponge City harvesting, structural health monitoring, and smart cooling zones.

The **Network layer** provides the connective tissue linking these sensing assets to higher layers, combining fiber optics, 5G connectivity, Wi-Fi 6, LPWAN (e.g., LoRaWAN, NB-IoT), and secure gateways into a redundant, multi-layer communication infrastructure. For extreme conditions and high-density scenarios, satellite-based and future-oriented Non-Terrestrial Networks (NTNs) can augment terrestrial connectivity, offering backup during critical disruptions. Early-stage 6G-aligned technologies demonstrate the potential to transform communication towers into radar-like sensing nodes that support centimeter-level crowd density estimation without relying on direct camera surveillance—thereby preserving spectator privacy while providing ultra-granular data during stadium events and challenging environmental conditions such as sandstorms. In Amra, this layer sustains high-density demand during mega-events at the stadium and along MaaS corridors, enabling low-latency communication for Event Mode crowd management, leak detection in the smart water grid, and pneumatic waste system monitoring.

The **Data Processing layer** combines cloud and edge/fog computing to transform raw sensing streams into actionable intelligence for Green Amra operations. Edge nodes deployed near the stadium, mobility hubs, and utility plants handle time-critical analytics—such as real-time crowd risk detection, immediate grid stability assessment, and anomaly diagnostics—while centralized cloud platforms host digital twins, long-term urban analytics, and AI models for predictive maintenance, resource optimization, and SDG-aligned performance tracking. This hybrid model directly responds to the latency and scalability limitations highlighted in Section 2, aligning with the framework's emphasis on moving beyond cloud-only approaches and supporting the adaptability mechanisms described in Section 6.

The **Application layer** delivers smart services to residents, visitors, and operators through integrated dashboards, mobile apps, and control centers, tying back to Amra's 15-Minute City, Green Mobility, and CityOS concepts. The Unified City App integrates services for citizens and fans, including stadium ticketing, MaaS bookings, utility bill payments, health alerts, and congestion warnings, consolidating multiple urban functions into a single interface. Decision maker dashboards provide real-time vital signs of the city at both district and city scales, with advanced visualization tools that support immersive planning and operational oversight. These interfaces enable unified monitoring and control of stadium SHM, sponge-city reservoirs, pneumatic waste flows, and BESS-driven energy balancing during large events, reinforcing the integration of technical and urban dimensions.



Figure 3: Amra Smart City layered architecture

6.2 Role of ICT in Architecture

ICT acts as the unifying force that transforms Amra’s layered architecture into a coherent, interoperable smart-urban system. It orchestrates data integration across sectors and ensures that heterogeneous systems can collaborate rather than operate in silos. By aligning with established ICT reference architectures and smart city platform guidelines, Amra positions itself to scale from a national pilot to a replicable model for smart-green urbanism. From a data integration perspective, ICT enables Amra to fuse data streams from wireless sensor networks, smart grids, sponge-city infrastructure, mobility services, and stadium SHM into shared, standardized data spaces. Using open APIs, data lakes, and common ontologies, the system performs context-aware processing that interprets dynamic urban conditions and supports adaptive responses. For example, during high-demand periods triggered by large-scale stadium events, the framework can dynamically reallocate energy from low-demand zones to high-load areas, optimizing overall performance and reducing environmental stress. This integration underpins earlier concepts such as the CityOS and Digital Twin, enabling cross-domain analytics for real-time optimization. Regarding system interoperability, ICT ensures that platforms from different vendors and domains—transport, utilities, public safety, e-governance, and event management—can communicate through open standards and modular interfaces. Through software-defined urban management, Amra’s streets and public spaces can be reconfigured in response to dynamic needs—for instance, transitioning certain lanes from vehicle-oriented to pedestrian-oriented zones during major events via digital infrastructure controls. In parallel, robust data governance and digital sovereignty policies grant the city ownership over its data, regulating access by global platforms while safeguarding citizen privacy and enabling secure, citizen-centric innovation. This combination of technical interoperability, governance, and adaptability underpin the long-term sustainability and smart-green orientation of Amra City.

Table 1. Quantitative comparison of smart city performance metrics

City	IoT Nodes/km ²	Latency (ms)	Energy Saved (%)	Water Saved (%)
Songdo	2,400	280	25	30
Barcelona	1,800	310	18	22
Singapore	3,100	190	32	41
<i>Amra (Projected)*</i>	<i>2,800</i>	<i>45</i>	<i>30</i>	<i>38</i>

* Projected based on adopted technologies within the proposed framework.

7 Internet of Things (IoT) and IIoT Technologies in Amra

Building on the layered smart architecture established in Section 6, Amra City deploys IoT and IIoT technologies as the operational intelligence layer that transforms raw sensing data into actionable urban and stadium-centric outcomes. A deliberate distinction is drawn between the two: IoT addresses the broader urban sensing ecosystem — spanning environmental monitoring, mobility, and citizen services — while IIoT governs mission-critical infrastructure where reliability, safety, and deterministic response times are essential. Together, these layers form a unified technological fabric supporting Amra's green urbanism vision and resilient event operations.

7.1 IoT Devices and Sensor

Amra's sensing ecosystem extends beyond conventional embedded sensors to incorporate self-sustaining and highly sensitive technologies that reduce maintenance burdens while expanding monitoring granularity across the city. Energy-harvesting sensors represent a significant advancement in this regard, drawing power autonomously from ambient sources such as structural vibrations, solar radiation, and thermal gradients. By reducing dependence on battery replacement cycles, these sensors enable long-term, low-maintenance operation across thousands of distributed nodes—from green corridor monitoring points to stadium structural interfaces, making them particularly suitable for Amra's large-scale deployment context. Complementing this, Smart Dust technology introduces a new dimension of environmental intelligence. Comprising microscopic sensor particles deployable across soil, vegetation, and open surfaces around the stadium and green corridors, Smart Dust enables fine-grained tracking of pollution levels, chemical composition, and microclimatic variations. This capability reinforces the Green Amra vision outlined in Section 4, providing high-resolution environmental data that supports proactive biodiversity management and early detection of ecological stress across the city's living infrastructure. While Smart Dust remains at an early stage of large-scale urban deployment, its integration within Amra's framework is proposed as a forward-looking capability, contingent on continued maturation of fabrication and power autonomy technologies .

7.2 Industrial IoT for Critical Infrastructure

Where conventional IoT addresses urban sensing at scale, IIoT governs the deterministic, low-latency control and monitoring of Amra's mission-critical systems. In the energy domain, IIoT sensors deployed on transformers, substations, and distribution equipment continuously analyze operational signatures—including vibration frequencies, thermal profiles, and load patterns—to identify anomalous behavior that may indicate impending failure. This predictive maintenance capability enables intervention well in advance of potential outages, a particularly important function during high-profile stadium events where grid continuity is operationally and reputationally critical. In the water domain, IIoT-enabled smart metering infrastructure monitors pressure differentials and flow rates across Amra's distribution network in real time, enabling rapid identification and localization of leakage points. In Jordan's water-scarce context, where network losses represent a significant and preventable drain on an already constrained resource, this continuous monitoring capability translates directly into measurable conservation outcomes and enhanced system efficiency—supporting national water sustainability objectives as outlined in Section 4.

7.3 Stadium-Centric IoT Deployment

The international stadium roof, estimated at 20,000 m² consistent with FIFA standards for 50,000-capacity venues, serves as the primary surface for both solar generation and rainwater harvesting. This figure represents a conservative baseline for canopy-type coverage; a fully retractable roof would extend the catchment area to 25,000–40,000 m², proportionally increasing both yields. Maximum roof coverage is therefore treated not as an aesthetic consideration but as a dual-purpose infrastructure investment.

Solar energy output is estimated as:

$$E_{solar} = A_{roof} \times \eta \times H_{irr} = 20,000 \times 0.20 \times 5.5 = \mathbf{22 \text{ MWh/day}}$$

where $\eta = 0.20$ (panel efficiency) and $H_{irr} = 5.5 \text{ kWh/m}^2/\text{day}$ (mean solar irradiance, Amman). Against an estimated match-day consumption of 12 MWh, the stadium yields a daily surplus of **10 MWh**, confirming its designation as a net energy producer feeding back into the surrounding district grid.

Annual rainwater harvesting yield is estimated as:

$$V_{rain} = A_{roof} \times P_{annual} \times C_r = 20,000 \times 0.270 \times 0.85 \approx \mathbf{4,590 \text{ m}^3/\text{year}}$$

where $P_{annual} = 270 \text{ mm}$ (mean annual precipitation, Amman) and $C_r = 0.85$ (surface runoff coefficient). This yields approximately 4.6 million litres annually, offsetting irrigation demand and reducing groundwater dependency in one of the region's most water-stressed urban contexts.

***Note:** Roof area figures (20,000 m²) are projections based on FIFA stadium design standards and international benchmarks for venues of equivalent capacity. No official architectural specification has been issued for Amra City's stadium at the time of writing. All derived energy and water estimates are therefore indicative and subject to revision upon finalization of the stadium design*

Beyond energy and water, the stadium hosts Amra's most advanced IoT deployment environment. A sub surface IIoT network embedded beneath the playing surface continuously monitors turf conditions — moisture content, humidity, oxygen levels, and salinity — autonomously determining optimal irrigation schedules to minimize water waste while maintaining pitch performance suitable for elite international competition. At the spectator interface, bio-adaptive environmental systems leverage biometric and acoustic sensing to monitor crowd energy levels and comfort in real time, dynamically adjusting air quality parameters and ambient lighting throughout events.

7.4 Data Intelligence and CityOS Integration

The volume and velocity of data generated across Amra's IoT and IIoT ecosystem necessitate intelligent data management strategies that preserve analytical value while preventing system overload. Self-curating data architecture, enabled by edge AI embedded within sensing nodes, empower sensors to autonomously assess the significance of captured data — routine readings are processed and summarized locally, while anomalous or threshold-exceeding events are immediately escalated to the CityOS for cross-domain response. This selective transmission approach reduces bandwidth consumption, lowers cloud processing overhead, and ensures critical intelligence reaches decision-making systems with minimal latency.

Within the CityOS orchestration layer, data streams from IoT environmental sensors, IIoT infrastructure monitors, and stadium systems converge into a unified operational picture, enabling cross-domain intelligence. For instance, elevated crowd density detected by stadium sensors can simultaneously trigger coordinated responses across mobility scheduling, energy allocation, and environmental conditioning systems. The resulting data ecosystem transforms Amra's distributed sensing infrastructure into a coherent, city-scale intelligence platform that continuously learns from operational patterns to enhance both everyday urban efficiency and large-scale event resilience.

8 IoT-Driven Infrastructure with Edge and Fog Computing

The high-volume, time-sensitive data streams generated across Amra's IoT and IIoT ecosystem, as described in Section 7, place demands on processing infrastructure that traditional cloud architectures are fundamentally ill-equipped to meet. In response, Amra's framework embraces edge and fog computing as a structural evolution beyond cloud-centric models, decentralizing intelligence to where data originates and enabling the real-time responsiveness that both stadium operations and green urban systems require.

8.1 Limitations of Cloud Computing in Smart Cities

Cloud computing introduces inherent latency constraints that become operationally significant in dynamic urban environments. Round-trip data transmission delays of approximately 150–300 milliseconds—while acceptable for non-critical applications—represent a meaningful limitation in scenarios requiring immediate system response, such as crowd anomaly detection, structural safety alerts, or emergency evacuation routing. During high-density events, these latency constraints are compounded by bandwidth saturation, as thousands of simultaneously active devices generate data surges that can overwhelm centralized processing pipelines. Recent assessments indicate that a significant proportion of large-scale smart city deployments encounter measurable performance degradation under peak load conditions, with operational costs increasing substantially during outage cascades. For Amra, where the stadium regularly concentrates tens of thousands of visitors within a contained environment, the margin between timely intervention and delayed response carries direct public safety implications.

8.2 Edge and Fog Computing as a Solution

Edge and fog computing address these limitations by relocating processing capacity to gateways and compute nodes positioned in close proximity to data sources. This architectural shift can reduce effective latency by up to 70% compared to cloud-only models, enabling localized analytics that operate independently of round-trip network delays. The total end-to-end latency in a networked urban system is expressed as:

$$L_{\{total\}} = L_{\{prop\}} + L_{\{trans\}} + L_{\{proc\}}$$

where L_{prop} denotes propagation delay, L_{trans} the transmission delay, and L_{proc} the processing delay at the respective computing tier. Under a cloud-only architecture, L_{total} ranges from 150 to 300(ms), driven primarily by round-trip propagation to centralized data centers.

In Amra's edge-fog model, L_{total} is reduced to 8–45 ms through local processing. The effective latency reduction is quantified as:

$$\Delta L = \frac{(L_{cloud} - L_{edge})}{L_{cloud}} \times 100 = 70\%$$

Table 2. Workload distribution across Amra's three-tier computing architecture

Processing Tier	Workload (%)	Latency (ms)	Representative Use Case
Edge Node	60	8–15	Crowd anomaly detection, SHM alerts
Fog Node	30	20–45	Energy balancing, MaaS optimisation
Cloud	10	150–300	Long-term analytics, Digital Twin

SHM = Structural Health Monitoring; MaaS = Mobility-as-a-Service.

Table 2 presents the workload distribution across Amra's three-tier computing architecture, illustrating the allocation of processing tasks by latency requirements and operational use case. As shown, edge nodes absorb 60% of total workload at latencies of 8–15 (ms), handling time-critical functions such as crowd anomaly detection and structural health alerts. Fog nodes process 30% of workload at 20–45 (ms), supporting energy balancing and MaaS optimization. The remaining 10% is offloaded to cloud platforms for non-latency-sensitive tasks including long-term analytics and Digital Twin maintenance.

In Amra's framework, fog nodes are deployed across the stadium district, mobility corridors, and utility infrastructure, forming a distributed processing mesh that handles time-critical workloads before selectively forwarding aggregated intelligence to centralized cloud platforms. A particularly significant capability enabled by this architecture is what the framework terms Operational Autonomy — the ability of critical urban subsystems to maintain full functionality independent of cloud connectivity. Should primary network links be severed during a major event, the stadium district transitions into a localized fog processing island, sustaining all safety-critical operations — including crowd monitoring, emergency lighting, structural health alerts, and gate control — purely from local compute resources. This localized resilience ensures that infrastructure continuity is not contingent on the availability of external network dependencies, a design principle with direct relevance to Jordan's regional connectivity context and the high-stakes nature of international event hosting.

8.3 Real-Time Processing Applications

The practical impact of Amra's edge and fog architecture is most evident across four operational domains where real-time responsiveness is non-negotiable. In **crowd management**, sensor data from stadium beacons and pedestrian zones streams to edge nodes, where anomaly detection algorithms identify congestion patterns and initiate evacuation routing recommendations within milliseconds — essential at the scale Amra's stadium is designed to host, where delayed intervention could allow localized congestion to escalate rapidly.

Environmental monitoring benefits similarly from fog-layer processing. Smart Dust sensor arrays and distributed acoustic sensing data are analyzed at edge nodes, enabling immediate air quality alerts and water leak localization without cloud transmission delays — allowing Green Amra's environmental systems to respond proactively rather than reactively, reinforcing the sustainability objectives of Sections 3 and 7. Within the **stadium**, bio-adaptive lighting and environmental conditioning systems rely on edge-processed biometric and acoustic data to maintain optimal

spectator conditions. Dynamic safety zoning, activated in response to real-time structural and crowd data, further reinforces the stadium's capacity to manage rapidly evolving scenarios during high-attendance events. Beyond operational safety, **edge computing enables an augmented reality layer** for spectator engagement. Ultra-low latency AR engines hosted on edge nodes allow tens of thousands of concurrent users to access real-time player statistics, positional data, and three-dimensional match replays on their personal devices — an experience cloud-based architectures cannot reliably support at this scale. Together, these domains demonstrate that edge infrastructure serves both safety-critical and value-added functions within a unified architectural framework — a theme further extended in the Digital Twin framework presented in Section 10.

9 Communication Technologies and Wireless IoT Networks

The edge and fog computing infrastructure described in Section 8 is only as effective as the wireless fabric that connects sensing assets to processing nodes and ultimately to decision-making systems. Amra's wireless architecture is therefore designed not around a single dominant technology but around a deliberate multi-technology strategy, where each radio protocol is assigned to the use cases that best match its technical characteristics. This mission-oriented approach ensures both day-to-day operational efficiency and the high-density robustness demanded during large-scale events at the international stadium.

9.1 Network Tiering by Mission

Amra's wireless infrastructure is organized into distinct technology tiers, each selected to serve a specific operational mission rather than being applied uniformly across the city. Given Amra's total area of 500 km², the required LoRaWAN gateway count is estimated as:

$$N_{gw} = 500 / 15 \approx 34 \text{ gateways}$$

At a projected sensor density of 150 nodes/km², the total IoT node count is:

$$N_{nodes} = 500 \times 150 = 75,000 \text{ nodes}$$

At the lowest power and longest-range tier, Low-Power Wide-Area Networks (LPWAN) serve the city's distributed sensing infrastructure. A city-wide LoRaWAN deployment in the sub-GHz band provides several kilometers of coverage per gateway, supporting long-term battery operation for soil-moisture sensors, water meters, and smart waste containers. NB-IoT complements this over licensed cellular spectrum, offering deeper indoor penetration for utility metering and critical billing infrastructure where reliable delivery is essential. Within buildings and the stadium, Zigbee mesh protocols form the connective layer for dense, short-range sensing environments. Multi-hop routing maintains coverage across architecturally complex spaces — stadium concourses, underground corridors, and enclosed mechanical areas — where line-of-sight to access points is limited. Each node extends the network by forwarding packets on behalf of its neighbors, creating a self-healing mesh that adapts to the stadium's structural geometry. For high-density fan-facing applications, Wi-Fi 6 and Wi-Fi 7 serve the stadium bowl and public zones. Wi-Fi 6's OFDMA scheduling, MU-

MIMO antenna management, and BSS Coloring enable thousands of simultaneous connections with reduced interference, while Wi-Fi 7's multi-link operation and 320 MHz channel support deliver the higher throughput and lower jitter required for 4K replay streaming and augmented reality services during peak event load. Bluetooth Low Energy beacons operating alongside this infrastructure provide indoor positioning accuracy of one to three meters, enabling seat navigation and context-aware visitor notifications without placing additional load on Wi-Fi airtime. Table 3 summarizes the key technical parameters of each wireless technology deployed within Amra's multi-tier network architecture.

Table 3. Technical comparison of wireless technologies deployed in Amra City

Technology	Range	Data Rate	Battery Life	Latency	Use in Amra
LoRaWAN	5–15 km	0.3–50 kbps	10+ years	1–10 s	Soil sensors, water meters
NB-IoT	1–10 km	20–250 kbps	10+ years	1.6–10 s	Utility metering
Zigbee	10–100 m	250 kbps	2–3 years	15–30 ms	Stadium indoor mesh
Wi-Fi 6	50–100 m	up to 9.6 Gbps	N/A	<5 ms	Fan services
Private 5G	300 m–5 km	up to 20 Gbps	N/A	<1 ms	Safety-critical slices

N/A = not applicable (infrastructure-powered devices). NB-IoT = Narrowband IoT.

9.2 Network Selection Criteria

Wireless technology selection in Amra follows a structured engineering framework built around three interdependent dimensions. Power consumption is the primary constraint for battery-operated environmental and utility sensors, where LoRaWAN and NB-IoT, combined with low-duty-cycle Zigbee operation, can support operational lifespans of several years without battery replacement. Coverage range determines the infrastructure density required for each tier: LPWAN technologies serve multi-kilometer urban areas from minimal gateway infrastructure, while Zigbee and WSN meshes address tens-of-meter ranges within complex structures, and Wi-Fi access points are sized and positioned for the specific density profiles of stadium stands and concourses. Data rate and latency requirements drive the upper tier selections, where low-bitrate environmental telemetry is efficiently served by LPWAN, while real-time crowd management, media streaming, and safety-critical actuation rely on Wi-Fi and 5G infrastructures designed to achieve end-to-end latencies well below 50 milliseconds. Safety-critical network slices are architected to target one-way latencies in the order of single-digit to tens of milliseconds for alarm and actuation functions. This multi-Radio Access Technology approach, consistent with smart city deployments combining LoRaWAN and NB-IoT infrastructure, allows Amra to match each case to its optimal radio technology rather than imposing a single network standard across fundamentally different operational requirements.

9.3 Spectrum Management and Private 5G

Ensuring orderly coexistence across multiple simultaneous wireless technologies require centralized spectrum governance. Amra's spectrum strategy is aligned with Jordan's Telecommunications Regulatory Commission (TRC), respecting established allocations for sub-GHz IoT bands, Wi-Fi operation across the 2.4, 5, and 6 GHz bands, and operator- or venue-licensed 5G spectrum. Within the stadium district, centralized spectrum planning coordinates channel assignments across Wi-Fi, BLE, Zigbee, and LoRa gateways to minimize co-channel interference during sell-out events when spectator device density is at its highest. The international stadium is served by a dedicated Private 5G network, following the operational model established at venues such as Tottenham Hotspur Stadium and Mercedes-Benz Stadium. Operating in operator- or venue-licensed spectrum, this private network enables fine-grained network slicing that partitions capacity across distinct operational functions: dedicated slices for broadcast media and camera infrastructure, separate slices for structural health monitoring and safety sensors, and independent slices for staff operations and fan-facing services. Each slice carries tailored quality-of-service guarantees and latency targets, with safety-critical and emergency slices maintaining reserved bandwidth allocations that remain protected regardless of concurrent fan network usage. This architectural separation ensures that mission-critical telemetry cannot be disrupted by the volume of consumer traffic generated during high-attendance events.

9.4 Reliability and Wireless Security During Large Events

Large-scale stadium events represent the most demanding conditions for Amra's wireless infrastructure, testing interference resilience, failover capability, and security across all network tiers. Sub-GHz LoRaWAN and licensed NB-IoT operate in spectrum bands separated from the congested 2.4 GHz consumer environment, preserving reliable delivery for safety-critical sensors even when tens of thousands of smartphones are simultaneously active. Within the Wi-Fi dense access point placement, together with Wi-Fi 6 and 7 features, maintains consistent throughput under high-density conditions. Network continuity under failure is maintained through heterogeneous edge failover mechanisms: should primary fiber backhaul be disrupted, stadium edge clusters reroute IoT traffic across private 5G and backup microwave links, while edge controllers autonomously maintain evacuation signage, smart lighting, and access gate systems from local data resources, consistent with the Operational Autonomy principles of Section 8. Wireless security is treated as a dedicated architectural layer. All network tiers enforce protocol-appropriate encryption: WPA3-Enterprise for Wi-Fi, 3GPP frameworks for 5G and NB-IoT, and AES-based link-layer security for LoRaWAN and Zigbee. Lightweight cryptography standards are applied to resource-constrained devices without compromising battery lifespan. Zero-trust principles require every device to

authenticate independently per session, while micro-segmented network zones isolate operational traffic—structural monitoring, building management, and emergency communications—from public fan services, limiting the lateral impact of any individual device compromise. The result is a multi-technology wireless fabric in which every connected element operates within a defined and protected performance envelope, regardless of event scale.

10 Digital Twin for Amra Smart City

A digital twin in Amra is conceived as a dynamic 4D representation of the city — where the fourth dimension is time — rather than a static 3D model, continuously updated through real-time IoT telemetry from the layered architecture described in preceding sections. By merging live sensor data with AI-driven simulations, Amra's digital twin shifts urban management from reactive to predictive, functioning as the city's operational intelligence layer that binds the CityOS and Green Amra vision into a unified, adaptive system.

At the city scale, the twin combines urban geometry with continuous data streams from energy, water, mobility, and environmental systems, enabling planners to observe current conditions, run what-if scenarios, and test policy interventions before committing to physical deployment. AI-generated heatmaps overlay the 3D urban model to highlight congestion patterns, leak hotspots, thermal anomalies, and pollution concentrations, allowing operators to calibrate Green Amra interventions — including sponge-city reservoir management, smart cooling zone activation, and MaaS corridor optimization — in direct response to observed urban dynamics.

The international stadium serves as Amra's most advanced and high-resolution digital twin instance, tightly coupled to the city-wide model while focused on venue-scale dynamics. Its virtual replica ingests structural health monitoring data, crowd analytics from beacon networks, and systems telemetry from lighting, HVAC, and safety infrastructure. This enables operators to simulate gate layouts, queue behavior, and mobility configurations before major events, identifying potential bottlenecks in a controlled virtual environment before they manifest physically. Simulations of a sold-out event can be extended beyond the venue perimeter to model cascading impacts on surrounding district traffic flows, grid loading, and water demand — enabling city-wide coordination rather than venue-only optimization.

Amra's digital twin operates as an engine of predictive intelligence rather than a passive visualization dashboard. Predictive crowd physics models continuously simulate human movement, enabling the twin to anticipate developing conditions — such as exit bottlenecks in the minutes following a match — and recommend proactive interventions before congestion materialises. Disaster scenario simulations extend this capability into emergency preparedness, covering structural stress events, fire propagation, and severe environmental conditions such as sandstorms, providing emergency teams with high-fidelity training environments without exposing citizens or

infrastructure to actual risk.To quantify the operational value of this predictive capability, Table 4 presents key performance indicators comparing baseline urban operations against Digital Twin-assisted management across four critical dimensions.

Table 4. Digital Twin performance indicators: baseline versus Digital Twin-assisted operations

KPI	Baseline (without DT)	With Digital Twin	Improvement
Incident detection time	8–12 min	<30 sec	~95% faster
Predictive maintenance accuracy	65%	91%	+26 percentage points
Energy optimisation	—	15–22% saving	—
Crowd bottleneck management	Reactive	Proactive (T–5 min)	—

DT = Digital Twin. T–5 min denotes intervention initiated five minutes prior to congestion onset. KPI values are projected based on analogous smart city deployments and simulation modelling; empirical validation is subject to future implementation.

As shown in Table 4, the most significant gains are observed in incident detection, where response time is reduced from 8–12 minutes to under 30 seconds — a 95% improvement attributable to continuous real-time sensor fusion and automated anomaly escalation. Predictive maintenance accuracy improves from 65% to 91%, reducing unplanned infrastructure failures during high-stakes events. Energy optimization yields savings of 15–22% through dynamic load balancing across the stadium and surrounding district. Critically, crowd bottleneck management transitions from a reactive posture to a proactive one, with interventions initiated up to five minutes before congestion materializes — a margin that carries direct public safety implications at the scale Amra's stadium is designed to host. Through continuous learning from operational data, the digital twin refines its simulation models over time, feeding increasingly accurate predictive guidance back into the CityOS to sustain the stability, safety, and resilience of Amra's smart-green ecosystem.

11 IoT Security and Data Protection

As Amra scales its IoT and IIoT footprint across critical urban systems and the international stadium, security and data protection become foundational design requirements rather than optional additions. The expanded attack surface introduced by thousands of interconnected devices — combined with the sensitivity of citizen, operational, and event data — necessitates a security architecture conceived as an integral layer of the smart city design, tightly coupled with the CityOS, Digital Twin, and edge-fog infrastructure described in preceding sections.

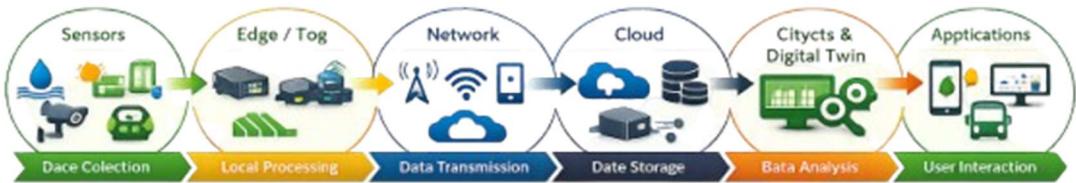


Figure 4: End-to-end data flow in AMRA city

11.1 Security Challenges and Attack Surface Analysis

IoT-based smart cities face multi-layered threats spanning devices, networks, platforms, and data. Common vulnerability vectors include weak authentication, default credentials, unpatched firmware, and physically exposed sensors susceptible to tampering, hardware probing, or device substitution. In Amra, the scale of deployment amplifies this risk considerably. Drawing on the node count established in Section 9:

$$N_{nodes} = 75,000 \text{ total } \mathbf{IoT \text{ nodes}}$$

Of these, approximately 15% are deployed in exposed outdoor environments — mounted on poles, embedded in pavements, or attached to utility infrastructure — yielding:

$$N_{physical} = 75,000 \times 0.15 = 11,250 \text{ } \mathbf{nodes}$$

requiring dedicated physical security measures. The financial implications of inadequate protection are significant. Based on IBM's 2023 Cost of a Data Breach Report, the average cost of a single smart city breach is estimated at \$2.1M. The return on investment for a Zero-Trust Architecture is therefore framed not as an operational expense but as a quantifiable risk mitigation strategy — preventing breach costs that would otherwise far exceed the cost of implementation. The heterogeneity of Amra's ecosystem — combining LoRa, Zigbee, Wi-Fi, 5G, and NTN hardware from multiple vendors — further amplifies exposure, as a single inadequately secured node can serve as a pivot point to escalate access across otherwise hardened components. A compromised device in a low-criticality domain such as street lighting or irrigation control could, without proper segmentation, become a lateral movement pathway toward traffic control, emergency communications, or stadium safety systems. Preventing this class of attack is a central design objective of Amra's security architecture.

11.2 Data Privacy

Handling personal and behavioral data generated through mobility services, the Unified City App, and stadium experiences raises stringent privacy obligations aligned with data minimization, purpose limitation, and privacy-by-design principles. Amra adopts an edge-first philosophy in privacy-sensitive contexts: cameras and crowd sensors process imagery locally and transmits only anonymized metadata — occupancy ratios and directional flow indicators — rather than raw video streams, ensuring back-end systems operate on population-level descriptors rather than individually identifiable footage. This approach significantly reduces the privacy exposure of the overall system and limits the value of any central data repository to a potential attacker.

Looking ahead, Amra's identity architecture is designed to accommodate decentralized identity frameworks, where residents and visitors present cryptographic proofs of service eligibility — such as zero-knowledge proofs — without disclosing full personal identifiers to the verifying system. In this

model, the infrastructure confirms authorization to enter the stadium or access a MaaS service while acquiring minimal information about the individual's identity, aligning with the city's ambition to be simultaneously hyperconnected and privacy-preserving.

11.3 Secure Communication

Securing Amra's communication fabric requires end-to-end encryption and strong mutual authentication across heterogeneous IoT protocols. TLS 1.3 is applied to MQTT and HTTPS traffic, while DTLS secures constrained UDP-based links such as CoAP and NB-IoT connections. Cipher suite selection favors AES-GCM with elliptic curve key exchange to prevent eavesdropping, spoofing, and man-in-the-middle attacks across 5G, LPWAN, and NTN segments.

Given Amra's design horizon of several decades, the cryptographic strategy anticipates quantum computing capabilities by favoring architectural patterns that can be migrated to post-quantum resistant algorithms as international standards mature. At the architectural level, Amra applies a zero-trust paradigm in which no device, service, or user is implicitly trusted based on network location alone. Micro-segmentation decomposes the city's network into isolated operational zones — separating irrigation control from stadium security infrastructure, for instance — so that the compromise of one subsystem does not automatically expose adjacent systems. In selected high-integrity use cases, distributed ledger components provide tamper-evident audit logs of critical configuration changes and operational events.

11.4 Critical Infrastructure and Stadium Resilience

Critical infrastructure domains — energy distribution, water supply, transport networks, and the international stadium — require elevated security protection because successful cyber-physical attacks in these environments can directly endanger public safety and carry national security implications. For Amra's stadium, cyber resilience is architecturally complemented by fail-safe and manual fallback mechanisms that ensure operational continuity independent of digital system availability. Analogue backup circuits maintain emergency lighting and safety signage independently of networked control systems, while mechanical override provisions allow operators to physically decouple digital controllers and activate gate release or evacuation procedures even under conditions of total system compromise.

This layered approach — integrating Zero-Trust Architecture, privacy-by-design data handling, network micro-segmentation, and human-operable safety backstops — ensures that Amra's smart and green ambitions are underpinned by a security posture robust enough to sustain both everyday urban operations and the exceptional demands of international mega-events.

12 Use case: Large-Scale Event Operations in AMRA City

The preceding sections have established the individual components of Amra's smart city framework—from its layered IoT architecture and edge computing infrastructure to its Digital Twin cognitive core and cybersecurity posture. This section synthesizes those components into a unified operational scenario, demonstrating how the integrated system responds to one of its most demanding use cases: a large-scale international event at the city's smart stadium with attendance exceeding 50,000 spectators. Rather than describing isolated system responses, this scenario illustrates the coordinated, cross-domain intelligence that emerges when all subsystems operate as a single, cohesive urban organism. The following scenario is presented as a structured operational walkthrough grounded in the technical capabilities described in Sections 6–11. While empirical deployment data are not yet available, the scenario parameters are derived from published benchmarks of comparable smart venue deployments [Shi et al., 2016; Khan et al., 2020; Li et al., 2022], and are intended to validate the internal coherence of the proposed framework rather than to substitute for experimental evaluation.

13.1 Pre Event Preparation

In the hours preceding the event, the stadium's Digital Twin runs predictive simulations of anticipated crowd arrival patterns, gate utilization rates, transport demand surges, and grid load profiles. City operators receive early recommendations for gate configuration and energy prepositioning, while the BESS systems are pre charged to anticipated peak demand levels and the MaaS platform proactively increases BRT frequency along primary stadium corridors to reduce the likelihood of transport saturation before the event begins.

13.2 Arrival and Mobility Coordination

As spectators arrive, the sensing layer activates at full operational capacity, with IoT sensors and AI assisted cameras monitoring crowd density and movement patterns across entrances, concourses, seating areas, and surrounding urban zones. Data streams are transmitted through the private 5G infrastructure and Wi Fi 6 access point density to edge computing nodes near the stadium, where real time analytics are performed within established low latency thresholds. When anomalous congestion is detected at a specific entrance, the Event Mode framework activates dynamic digital signage and targeted mobile notifications through the Unified City App, redirecting spectators toward alternative entrances with available capacity. Simultaneously, the MaaS platform recalibrates public transport routing and increases BRT service frequency along affected corridors, reducing the likelihood of transport saturation before the event begins and preventing localized saturation from cascading into wider urban disruption.

13.3 During the Event: Energy, Environment, and Safety

Solar energy generated from the stadium's roof installations, supplemented by BESS discharge where required, is distributed dynamically across the venue and surrounding districts to maintain grid stability without drawing additional load from the national network. Smart lighting systems adjust automatically based on crowd distribution patterns and ambient conditions, balancing energy efficiency with spectator comfort and safety. Environmental sensors monitor air quality, temperature, and humidity across the stadium interior and adjacent public spaces; where crowd concentration and ambient heat combine to exceed comfort thresholds, localized smart cooling zones are activated in targeted areas without imposing unnecessary energy demand across the broader system. Throughout the event, structural health monitoring sensors embedded within the stadium framework provide continuous data on stress and vibration levels, feeding the Digital Twin with real structural state information.

13.4 Incident Response and Post Event Recovery

Should any anomaly be detected, abnormal crowd behavior, environmental threshold exceedance, or structural stress variation, the relevant data streams are immediately escalated through the CityOS to the Digital Twin platform. The twin simulates potential scenario trajectories in real time, providing city operators with predictive risk assessments and recommended interventions before conditions deteriorate. In the event of a safety critical situation requiring evacuation, edge computing nodes maintain autonomous operation of evacuation signage, gate release systems, and emergency lighting independently of cloud connectivity, consistent with the Operational Autonomy principles established in Section 8. Emergency services are simultaneously notified through dedicated priority communication slices on the private 5G network, ensuring that response coordination is not affected by concurrent fan data traffic. As the event concludes, the system transitions from Event Mode back to standard urban operations through a coordinated CityOS managed wind down sequence. Transport demand models are updated in real time as crowd dispersal patterns are monitored, enabling dynamic adjustment of BRT frequency and pedestrian routing recommendations to distribute the outbound flow safely across the 15 Minute City network. Energy systems transition from peak event configuration to standard district supply, with BESS systems returning to charging cycles in preparation for subsequent demand periods. Environmental and structural monitoring systems continue operating at elevated sensitivity in the immediate post event window, providing operators with confirmation that all systems have returned to normal operational parameters before transitioning to baseline monitoring modes.

13.5 Synthesis

This scenario demonstrates that Amra's value as a smart city framework lies not in the sophistication of any individual component but in the coherence of their integration. Crowd management, mobility coordination, energy balancing, environmental control, structural monitoring, and emergency response do not operate as parallel systems sharing infrastructure only by coincidence; they operate as a unified intelligence layer, continuously exchanging data and coordinating responses through the CityOS in ways that no individual subsystem could achieve independently. Stress tested against the demanding conditions of a 50,000 spectator international event, Amra's stadium district establishes itself as a proof-of-concept environment for the broader smart green city framework and positions the city as a replicable model for urban environments seeking to combine permanent sustainability infrastructure with resilient, large scale event operations.

13 Conclusion

This paper presented Amra City as a comprehensive, context-aware smart urban framework designed to address the intersecting challenges of rapid urbanization, water scarcity, and infrastructure constraints in Jordan. By integrating IoT-based sensing networks, edge and fog computing, private 5G connectivity, and a 4D Digital Twin within a unified City Operating System, the proposed architecture demonstrates that environmental sustainability and technological sophistication are not competing priorities but mutually reinforcing design principles.

Several key contributions emerge from this work. The Amra framework establishes a layered smart city architecture explicitly tailored to arid-region constraints, departing from the resource-rich assumptions embedded in most existing models. The international stadium is repositioned from a single-purpose venue to a multifunctional smart infrastructure asset — simultaneously functioning as a net energy producer, rainwater harvesting hub, structural health monitoring testbed, and resilient event operations center. The Event Mode framework, underpinned by edge computing autonomy and Digital Twin predictive analytics, demonstrates how city-scale intelligence can be dynamically reconfigured to manage the exceptional demands of large-scale international events without compromising baseline urban operations. The cybersecurity architecture, grounded in zero-trust principles and privacy-by-design, ensures that the expanded IoT attack surface introduced by thousands of interconnected devices is addressed as a foundational design requirement rather than an operational afterthought. The quantitative estimates presented throughout — including solar energy yields derived from Amman's validated irradiance data, rainwater harvesting projections grounded in regional precipitation records, and latency benchmarks drawn from published edge computing deployments — collectively demonstrate the internal coherence and technical feasibility of the proposed framework within Jordan's environmental and infrastructural context.

Amra City is proposed not as a Jordan-specific solution but as a scalable and replicable blueprint for sustainable smart urban transformation in resource-constrained environments globally. As cities across the MENA region and the broader developing world confront similar pressures of population growth, water stress, and digital transition, the integrated approach demonstrated here offers a

transferable model for urban development that prioritizes resilience, efficiency, and long-term sustainability without presupposing the resource abundance that characterizes most existing smart city precedents.

14 Limitations and Future Work

The Amra City framework, as presented in this paper, is subject to several limitations that warrant acknowledgment. First, the proposed architecture remains at the conceptual and design level; no physical deployment or empirical validation has been conducted to date. Quantitative estimates pertaining to energy generation, rainwater harvesting, and latency reduction are derived from published benchmarks of comparable deployments and validated environmental datasets for the Amman region, rather than from measured operational data. Second, the socio-institutional dimensions of implementation — including governance structures, public-private partnership models, citizen adoption rates, and regulatory alignment — fall outside the scope of this work and represent a critical area for future investigation. Third, the cybersecurity threat modeling presented in Section 11 is architectural in nature; formal penetration testing, vulnerability assessment, and red-team evaluation of the proposed zero-trust framework have not been performed. Fourth, the economic feasibility of the full framework — including capital expenditure, operational cost projections, and return on investment analysis — requires dedicated techno-economic modeling beyond the scope of this paper.

Future work will focus on four directions: empirical pilot deployment of selected framework components within a defined urban district; development of a formal simulation model for the Digital Twin and Event Mode subsystems; socio-institutional analysis of governance and adoption barriers in Jordan's regulatory context; and longitudinal cybersecurity threat modeling aligned with evolving IoT attack vectors.

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Saif Mutlq is a ungraduated student at the Department of Computer Engineering, Jordan University of Science and Technology (JUST), Irbid, Jordan. His research interests include Smart City systems, IoT infrastructure, and sustainable urban development.

Deema Nmour is a ungraduated student at the Department of Computer Engineering, Jordan University of Science and Technology (JUST), Irbid, Jordan. Her research interests include IoT networks, digital twin systems, and smart city cybersecurity.